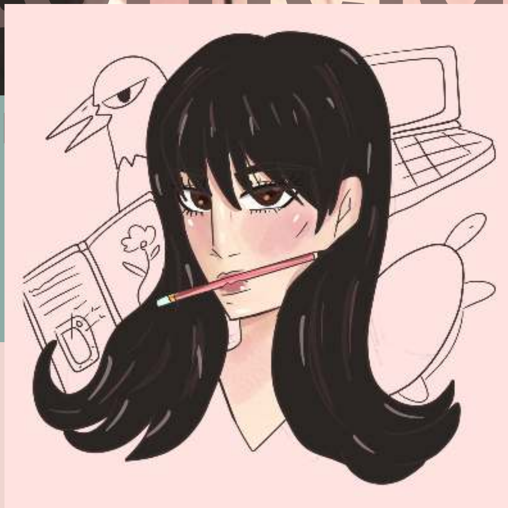


(PROBABLY ALMOST)

# EVERYTHING ON THE SYLLABUS

## XBI11 IAL EDEXCEL BIOLOGY

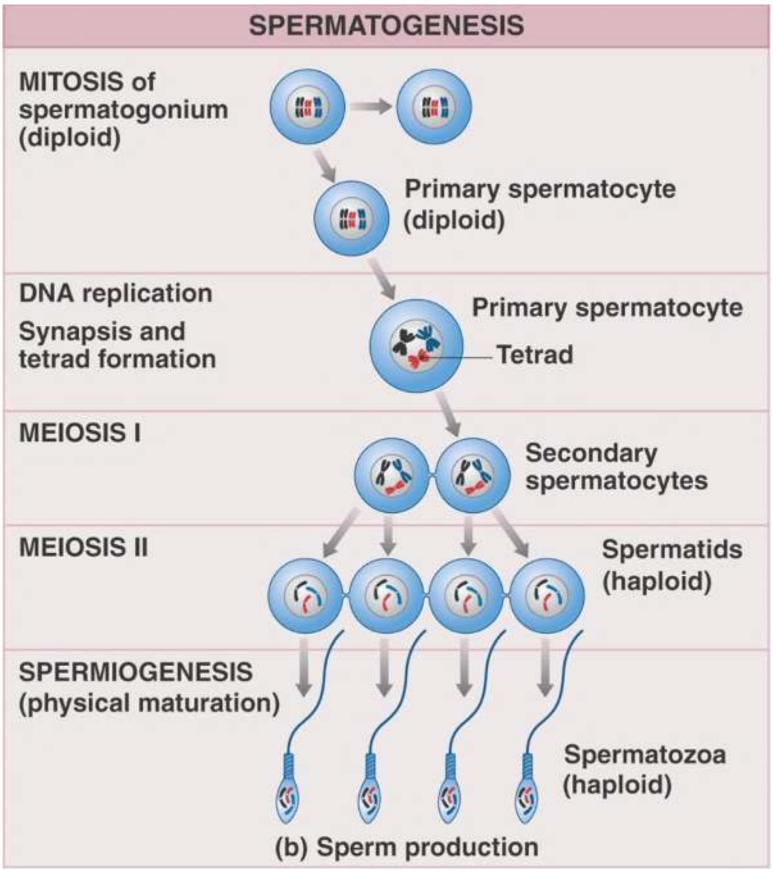
### UNIT 1 & 2

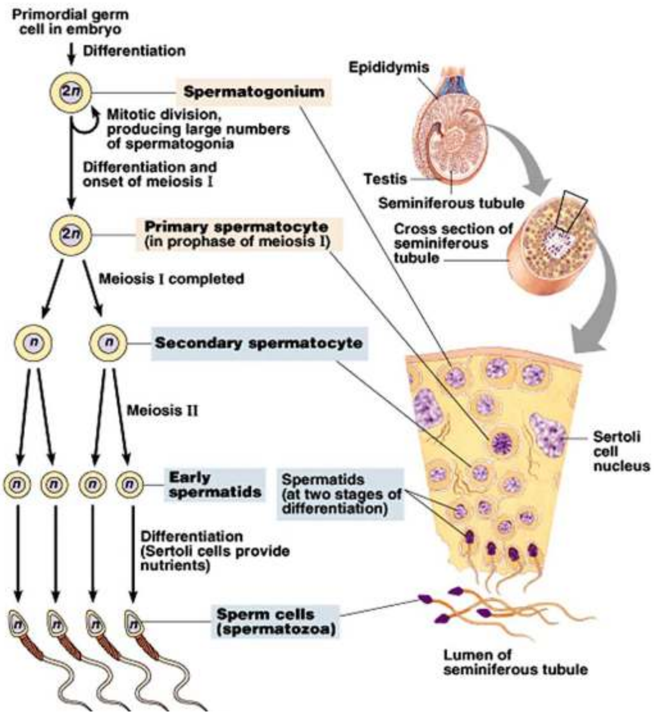
# Reproduction

## Reproduction in Mammals

Gametogenesis

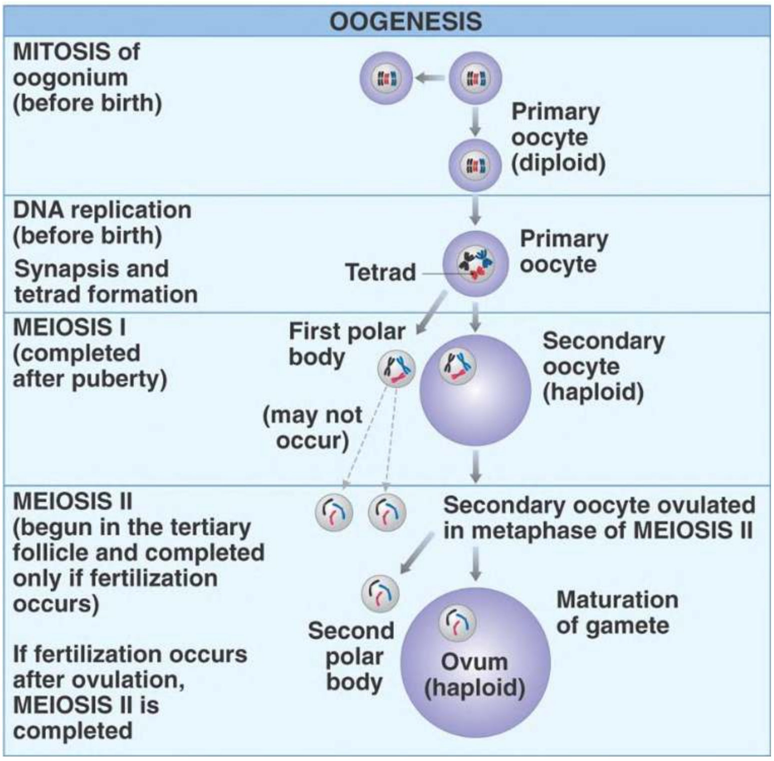
▼ Describe the Stages of Spermatogenesis

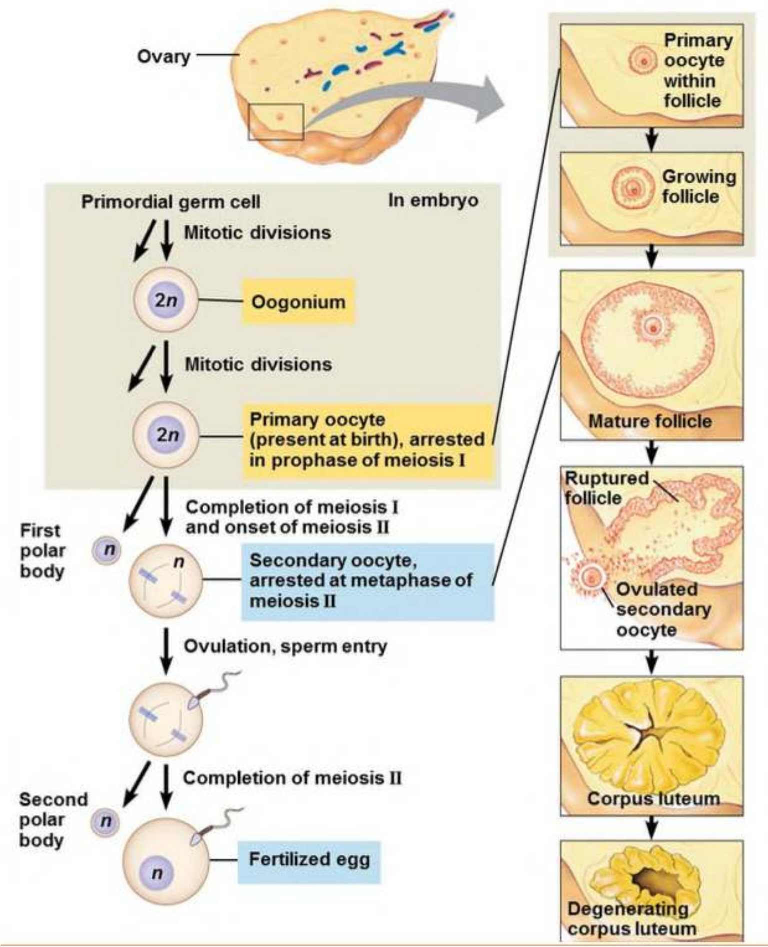





- Priomordial germs cells in the embryo undergo differentiation to form spermatogonium.
- Spermatogonium divides by mitosis multiple times to produce many diploid spermatogonia stem cells.
- One of these spermatogonia **differentiate** and undergoes **Prophase of meiosis 1** *(DNA replication, synapsis/crossing over and tetrad formation)*. At the end of P1, a **diploid Primary Spermatocyte cell** is formed.
- The Primary Spermatocyte then completes **meiosis** 1 - after the **first** meiotic division, the primary spermatocyte forms **2 *haploid* secondary spermatocytes.**
- Second meiotic division then forms **4 haploid spermatids** which differentiate into mature **spermatozoa/sperm cells**

▼ Describe the stages of Oogenesis


OOGENESIS
MITOSIS of oogonium (before birth)
Primary oocyte (diploid)
DNA replication (before birth)
Synapsis and tetrad formation
Tetrad
Primary oocyte
MEIOSIS I (completed after puberty)
First polar body
(may not occur)
Secondary oocyte (haploid)
MEIOSIS II (begun in the tertiary follicle and completed only if fertilization occurs)
Secondary oocyte ovulated in metaphase of MEIOSIS II
Second polar body
Maturation of gamete
Ovum (haploid)
If fertilization occurs after ovulation, MEIOSIS II is completed


Ovary
Primordial germ cell
In embryo
Mitotic divisions
2n
Oogonium
Mitotic divisions
2n
Primary oocyte (present at birth), arrested in prophase of meiosis I
Completion of meiosis I and onset of meiosis II
First polar body
n
n
Secondary oocyte, arrested at metaphase of meiosis II
Ovulation, sperm entry
Completion of meiosis II
Second polar body
n
n
Fertilized egg
Primary oocyte within follicle
Growing follicle
Mature follicle
Ruptured follicle
Ovulated secondary oocyte
Corpus luteum
Degenerating corpus luteum

Before Birth:

- Primordial germ cell in embryo undergo mitosis to form Diploid Oogonium cell.
- The Oogonium cell thend ivides by mitosis to form primary oocyte, and then is arrested in prophase of meiosis 1 *(after DNA replication, synapsis/crossing over and tetrad formation)*

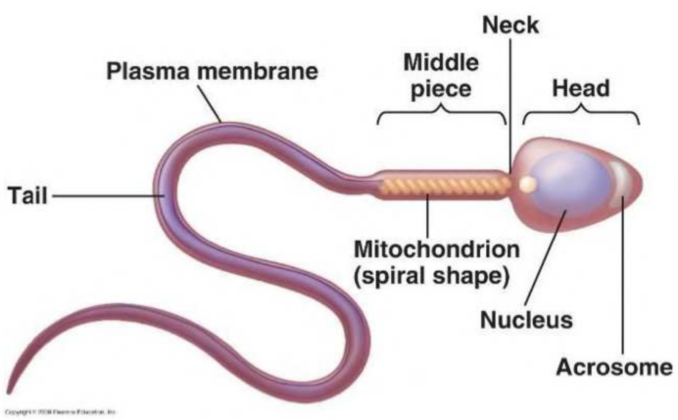
At Puberty:

- Meiosis I is completed to produce the **first polar body** and the **haploid secondary oocyte** is produced.
- **The secondary oocyte continues to meiosis II and is arrested at metaphase of meiosis II**

At Fertilisation (ovulation, sperm entry):

- Meiosis II is completed to form the second polar body and a haploid ovum/fertilised egg.

▼ Outline how Sperm Cells are specialised for their function



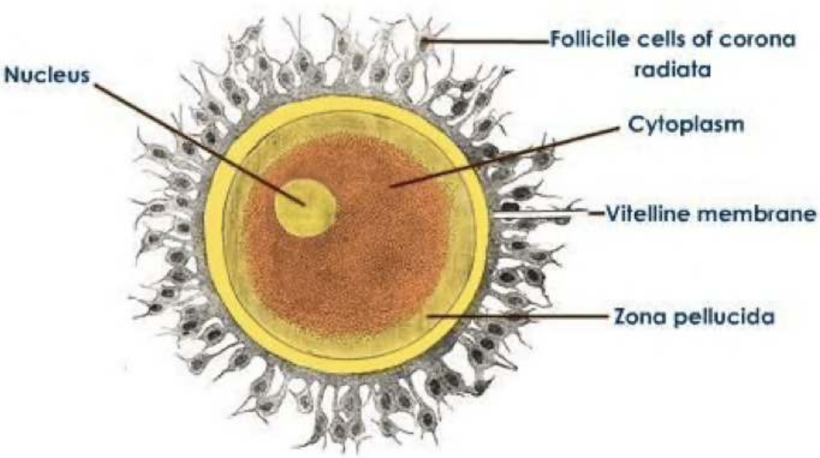

Features : Functions

1. Lots of spiral **mitochondria**: **Provides energy for movement** of the flagellum
2. **Flagellum**/Tail: Locomotion/movement. Enables the sperm cell to move and **swim to the egg cell** to fertilise it.
3. **Acrosome**: large lysosome containing **digestive/acrocin enzymes** which **break down the zona pellucida** (protective coating of egg cell)

and allow sperm to **penetrate the egg.**

4. **Haploid Nucleus:** Contains the genetic information from father. Haploid so that a full set of chromosomes are restored at fertilisation.

▼ Outline how Egg Cells are specialised for their function




Features : Functions

1. **Zona Pellucida**: Protective coating which the sperm has to penetrate for fertilisation to occur. Prevents more than one sperm fertilising the egg.
2. **Haploid Nucleus:** Contains the genetic information from mother. Haploid so that a full set of chromosomes is restored at fertilisation.
3. **Cortical Granules: Secretory vesicle/lysosome** that fuses with the egg cell membrane and releases an enzyme (via exocytosis) which **causes the zona pellucida to harden** to **prevent more than one sperm cell fertilising the egg (polyspermy).**
4. **Lipid Droplets**: Provides the energy needed for the developing zygote
5. **Follicle Cells**: Form a **protective coating** around the egg

▼ Outline the Acrosome Reaction

Acrosome, a lysosome, releases acrosin enzyme which digests the zona pellucida. The sperm cell can then penetrate the zona pellucida and fuse with the secondary oocyte membrane to allow the sperm nucleus to enter the egg cell.

▼ Outline the Cortical Reaction

Upon fertilisation, cortical granules that are lysosomes fuse with the egg cell membrane and release an enzyme via exocytosis. This enables the cortical reaction where the zona pellucida hardens, preventing polyspermy

Fertilisation

▼ Describe the process of fertilisation in mammals

Fertilization is the process in which a haploid female gamete (oocyte/egg cell) and a haploid male gamete (sperm cell) fuse and combine genetic material to form a single diploid nucleus known as zygote.

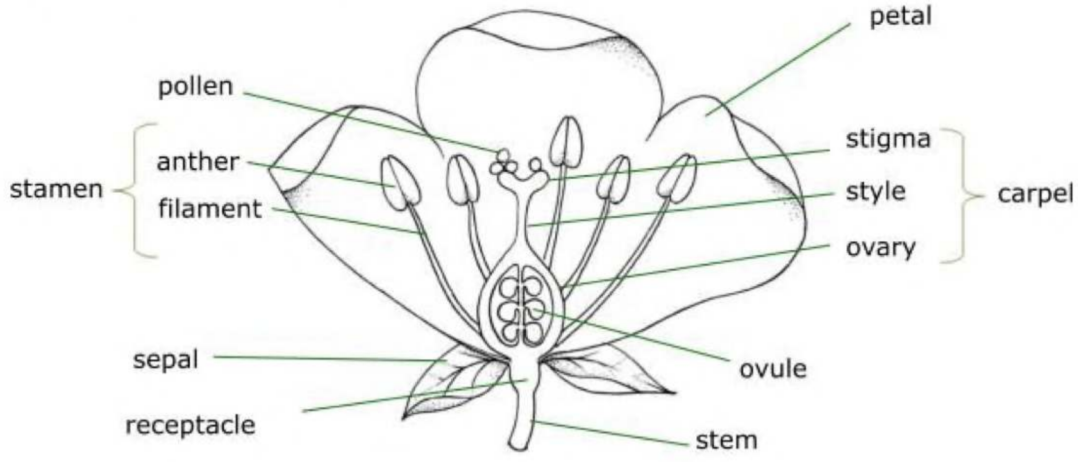
Once a sperm cell enters the female reproductive tract, it undergoes **capacitation** which changes the sperm cell membrane and increases motility.

1. Once the sperm head meets the **zona pellucida** (protective jelly layer), the **acrosome reaction** occurs. The acrosome bursts and causes digestive acrosin enzyme to move out of the acrosome and start digesting the zona pellucida.
2. The sperm membrane then **fuses** with the secondary oocyte membrane which allows the sperm nucleus to enter the egg cell.
3. The **cortical reaction** occurs where cortical granules are released from the oocyte. The cortical granules are lysosomes which fuse with the egg cell membrane and release enzymes by exocytosis which causes the zona pellucida to harden to **prevent polyspermy.**
4. *Sperm entry stimulates the completion of meiosis II of the secondary oocyte to produce the ovum and second polar body*.
5. **The sperm nucleus and oocyte nucleus fuse, producing a diploid zygote such that the full set of chromosomes is restored.**

## Reproduction in Angiosperms

▼ What are different structures in an angiosperm

## Structure of flowers



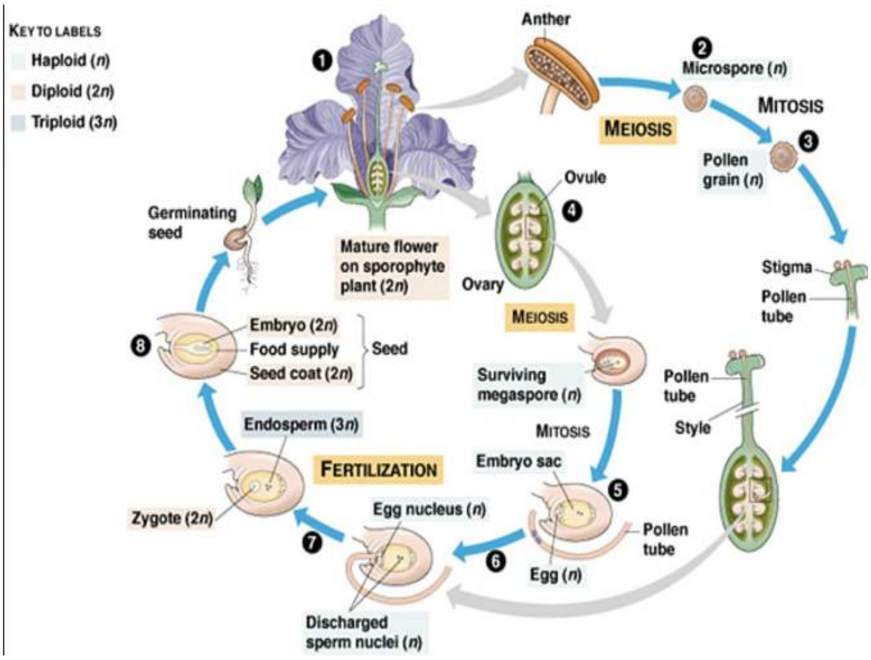

1. Male Reproductive Parts: **Stamen**, which consists of the long filaments and anthers.
    - **Filament**: Stalk that supports the anther
    - **Anthers**: Produces male gametes in the form of pollen grains.
2. Female Reproductive Parts: Carpel/Pistil, which consists of the stigma, the style and the ovule.
    - (Sticky) **Stigma**: Site at which pollen grains sticks /adhere in order to start the fertilisation process.
    - **Style**: Stalk that supports the stigma
    - **Ovule**/Ovary: Site of fertilisation, develops into fruit after fertilisation.
3. Non-gender Specific Parts: Sepal and Petal
    - **Sepal**: Provide protection for developing flower buds
    - **Petals**: Are colourful which attracts pollinators.

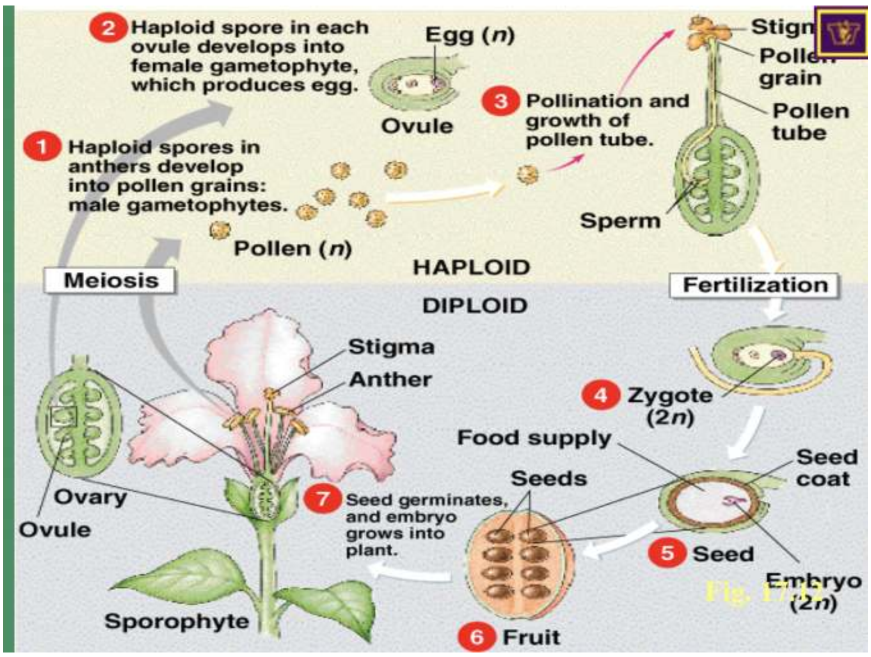
▼ What is pollination?

The process by which pollen grains (male gamete) produced by anthers are transferred to the female reproductive organs (stigma) of a plant.

Occurs by wind or through pollinators.

▼ Describe the process of fertilisation in flowering plants.




Mature pollen grains consist of two types of cells: a tube cell and a generative cell

1. Mature Pollen grains consisting of the pollen tube cell and the generative cell adheres to the stigma and germinates to produce a pollen tube that grows down to the style.
2. The tip of the pollen tube produces hydrolytic enzymes which digests the tissue of the style, creating a pathway that allows the pollen tube to

grow down to the ovary. (The digested tissue acts as a nutrient source for the pollen tube.)

3. The generative cell of the pollen **divides** to produce two male nuclei which later enter the embryo sac.
4. The pollen tube grows through a gap (between the integuments), known as the **microphyle** of the ovule and into the **embryo sac.**
5. One of the male gametes **fuses** with the female gamete/nucleus (egg cell) to form a **zygote.** The other male gamete fuses with the **two polar nuclei** to form the **endosperm (triploid)** which serves as a source of nutrients to the developing embryo.
6. At this point, fertilisation is complete. The fertilised ovule divides by mitosis to form  the embryo consisting of the developing shoot known as the **plumule**, developing root known as the **radicle** and one or two **cotyledons**. The integuments become the seed coat, the ovule becomes the seed and the ovary becomes the fruit.

**This PDF is a Compilation of Notes I (@attorneighhh) Created in my AS Year. It contains information from a wide range of sites and textbooks, including, but not limited to:**

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